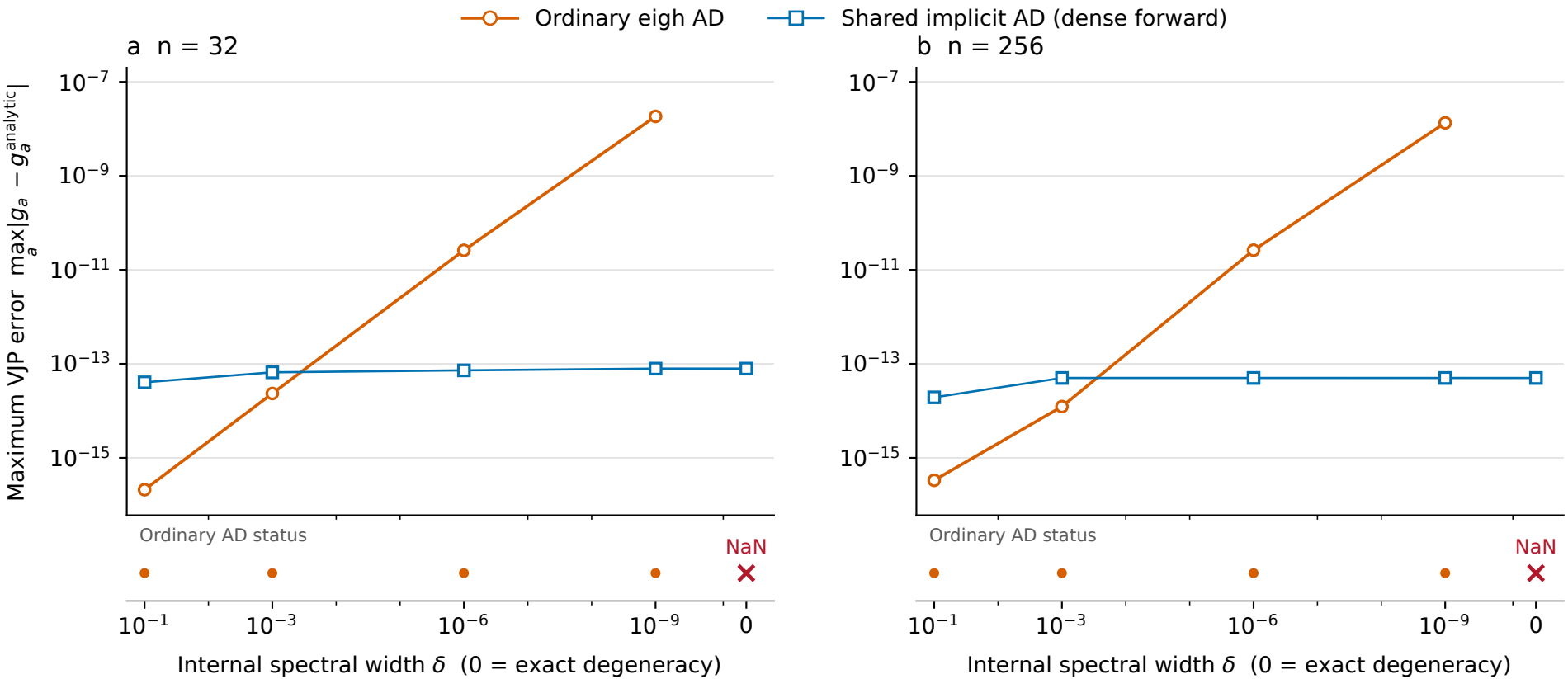
Same smooth projector loss: ordinary eigh AD fails numerically at exact degeneracy
$L = \langle W, B^{\mathsf{T}} P B \rangle$ · complete rank-4 cluster · fixed exterior gap $g = 0.7$



Exact-degeneracy forward check (dimensionless):
n=32: loss error 2.22e-16, minimum raw internal gap 0.0e+00; n=256: loss error 4.72e-16, minimum raw internal gap 0.0e+00

Nonfinite ordinary VJPs appear only as red crosses in the separate status strip, never as zero errors.
Shared analytic-reference gates pass for every case. Observed with JAX 0.10.0, CPU float64; this is an implementation result, not mathematical nondifferentiability.